Leveraging LLMs for Video Querying

submitted in partial fulfillment of the requirement
for the award of the Degree of

Bachelor of Technology
in
Computer Engineering

by

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May 2023

Certificate

This is to certify that the Project entitled "Leveraging LLMs for Video Querying" has been completed to our satisfaction by Mr. Deep Nayak, Mr. Atharva Mohite and Mr. Ameya Jangam under the guidance of Dr. Anant Nimkar for the award of Degree of Bachelor of Technology in Computer Engineering from University of Mumbai.

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Seal of the Institute

Statement by the Candidates

We wish to state that the work embodied in this thesis titled "Leveraging LLMs for Video Querying" forms our own contribution to the work carried out under the guidance of Dr. Anant Nimkar at the Sardar Patel Institute of Technology. We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

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Abstract

This research paper explores the application of Large Language Models (LLMs) in enhancing video search capabilities. The focus is on utilizing the Bi-modal Transformer (BMT) and Vid2Seq models for video captioning, using the UCF Crime dataset for experimentation. Various LLMs, including GPT3.5, GPT4, and LLAMA, are employed to create an efficient search system that can precisely pinpoint specific moments within videos. The methodology involves generating text-based captions using the BMT and Vid2Seq models, conducting experiments with different LLMs and prompts, and evaluating system performance.

The results demonstrate the transformative potential of LLMs in advancing video captioning and revolutionizing video search capabilities. By leveraging advanced language models, the accuracy and efficiency of the video search system are significantly improved. The findings highlight the importance of prompt design and model selection in achieving optimal results.

This research makes a valuable contribution to the field of video search by exploring the potential of LLMs in improving search efficiency. By harnessing the power of advanced language models and training them on diverse datasets, the study showcases their ability to enhance video captioning and revolutionize the way users interact with video content. The insights gained from this research provide a solid foundation for further advancements in video querying techniques and contribute to the ongoing efforts to enhance video search capabilities in an increasingly visual digital landscape.

Chapter 1

Introduction

Efficiently searching and pinpointing specific moments within videos is of paramount importance, particularly in the context of crime videos. This paper delves into the application of Large Language Models (LLMs) to enhance video search capabilities, with a specific focus on querying crime-related videos. The study leverages the Bi-modal Transformer (BMT) model, which is fine-tuned on a handcrafted caption dataset tailored for crime videos, resulting in a robust and contextually relevant video captioning system.

Existing video search systems often struggle with accurately identifying and retrieving relevant moments within crime videos. By harnessing the power of LLMs and fine-tuning the BMT model on a handcrafted caption dataset specifically designed for crime videos, we aim to address this challenge. This approach allows us to develop an efficient search system that can precisely identify and retrieve crucial moments related to criminal activities.

In the state of the art, the application of Large Language Models (LLMs) for video querying, particularly in the context of crime videos, remains relatively unexplored. The capabilities of LLMs like GPT3.5, GPT4, and LLAMA have been demonstrated in various natural language processing tasks, but further investigation is needed to determine their effectiveness in enhancing video search capabilities. This research gap is addressed in this paper, which aims to explore the potential of LLMs when attached on top of the Bi-modal Transformer (BMT) and Vid2Seq models, for efficient and precise video querying. The captions generated by the BMT and Vid2Seq models are queried using these LLMs, enabling improved video search capabilities in the domain of crime videos.

The proposed solution involves fine-tuning the BMT model on a carefully curated caption dataset consisting of crime videos. This dataset encompasses various criminal activities, allowing the model to capture the nuances and specificities of crime-related context. By training the BMT model on this dataset, its ability to generate accurate and relevant captions tailored to the crime domain is enhanced.

To evaluate the performance of the system and measure the accuracy of event occurrence in crime videos, experiments are conducted using different LLMs, including GPT3.5, GPT4, and LLAMA. By comparing their outputs and analyzing the impact of various prompts, insights are gained into the strengths and limitations of LLM-based video querying techniques in the crime domain.

1.1 Motivation

The motivation for this project stems from the ever-increasing volume of video content available online and the need for efficient and accurate video search capabilities. With the proliferation of platforms like YouTube, TikTok, and social media, users are faced with the daunting task of sifting through vast amounts of video content to find specific information or moments of interest. Traditional methods of video search, such as metadata tags or manual annotations, often fall short in capturing the rich and complex nature of video content. Therefore, the researchers aim to explore the potential of Large Language Models (LLMs) in enhancing video search by leveraging their ability to understand and generate natural language. By developing a more advanced video search system that can accurately and efficiently pinpoint specific moments within videos, this research seeks to empower users to navigate and interact with video content more effectively. Ultimately, the motivation behind this project lies in revolutionizing the way users discover and engage with videos, thereby addressing the growing demands of the visual digital landscape.

1.2 Problem Statement

To addresses the need for efficient and accurate video search capabilities by exploring the application of Large Language Models (LLMs) in enhancing video captioning and revolutionizing video search, with a focus on prompt design, model selection, and system performance evaluation

1.3 Objectives

- Develop and implement a video search system utilizing Large Language Models (LLMs) for accurate and efficient video captioning.
- Experiment with different LLMs, including GPT3.5, GPT4, and LLAMA, to evaluate their performance and effectiveness in enhancing video search capabilities.
- Investigate the impact of prompt design on the accuracy and efficiency of the video search system powered by LLMs.
- Explore the potential of LLMs trained on diverse datasets, such as the UCF Crime dataset, to enhance video querying techniques and revolutionize the way users interact with video content.

1.4 Scope

- Application of Large Language Models (LLMs) in enhancing video search capabilities, specifically focusing on video captioning and system performance evaluation.

- Evaluation of different LLMs, such as GPT3.5, GPT4, and LLAMA, fall within the scope of this research to assess their impact on improving the accuracy and efficiency of the video search system.
- Exploring the influence of prompt design on the performance of LLM-based video search, with an emphasis on achieving optimal results in terms of accuracy and search efficiency.

Chapter 2

Literature Survey

2.1 Video Captioning

For as long as computer vision has been, crime detection has been one of the field's toughest problems to solve.

Video captioning models [4] are deep learning models designed to automatically generate textual descriptions or captions for videos. These models aim to bridge the gap between visual content and natural language understanding by analyzing the visual information within videos and generating corresponding textual descriptions. Video captioning models typically employ a two-stage approach. In the first stage, a video encoder processes the input video frames or clips, extracting visual features that capture the temporal and spatial information present in the video.[13] These features encode the visual context necessary for generating accurate captions. [5] In the second stage, a language decoder takes the encoded video features and generates the textual captions. This decoder can be based on recurrent neural networks (RNNs), convolutional neural networks (CNNs), or transformer models, depending on the specific architecture used. The decoder generates the captions by sequentially predicting words or phrases[10], conditioned on the visual features and previously generated text. [7]. To train video captioning models, large-scale datasets containing videos paired with human-annotated captions are essential. These datasets provide the necessary supervision for the models to learn the mapping between visual content and textual descriptions. The training process involves optimizing the model's parameters to minimize the discrepancy between the generated captions and the ground truth captions.

Bi-modal Transformers, a vision-language model based on transformers, have garnered significant attention and demonstrated remarkable strengths in the field of video querying. These models effectively capture complex relationships within and between visual and textual representations, enabling robust performance in tasks like visual question answering (VQA) and bi-directional image and text generation. The integration of the Contrastive Language Image Pretraining (CLIP) network facilitates the seamless embedding of image patches and question words, while attention mechanisms adeptly capture dependencies. The averaging of predictions from classifiers mounted on top of contextual representations yields reliable final answers. Moreover, the adoption of a unified multimodal framework based on the Transformer architecture showcases the flexibility and adaptability of Bi-modal Transformers in generating coherent sequences of images and text. Bi-modal Transformers exhibit remarkable strengths in video querying, capturing intricate relationships within visual and textual representations. Nonetheless, it is crucial to recognize their limitations, such as potential biases stemming from training data, computational complexity, and performance variations across domains and datasets. To overcome these challenges and bolster the applicability and reliability of Bi-modal Transformers in video querying, addressing these weaknesses necessitates meticulous attention to fair training data curation, optimization of computational efficiency, and robust generalization across diverse contexts. By addressing these factors, the potential of Bi-modal Transformers can be fully harnessed for enhanced video querying capabilities.

2.2 LLMs

Large Language Models (LLMs) [3] are advanced deep learning models that have been trained on vast amounts of text data to understand and generate human-like language. These models utilize sophisticated neural network architectures, such as transformers, to process and generate text based on the patterns and structures they have learned from the training data. Large Language Models (LLMs) have revolutionized the field of language modeling, leveraging advanced deep learning techniques and transformer architectures to understand and generate human-like language. Prominent examples of LLMs include models like GPT 3.5, GPT4, and LLaMA. These models showcase substantial improvements in language understanding and generation capabilities, surpassing their predecessors and achieving competitive performance on various natural language processing benchmarks. By training on extensive and diverse text corpora, these LLMs capture intricate patterns and structures, enabling them to generate contextually rich and coherent text. The incorporation of self-attention mechanisms and multi-layered feed-forward networks further enhances their ability to process and comprehend language. These advancements in LLMs pave the way for a wide range of applications in areas such as information retrieval, text generation, and natural language understanding, driving the progress of language processing technologies. Large Language Models (LLMs), including GPT 3.5, GPT4, and LLaMA, represent notable strides in language understanding and generation. These models excel in capturing intricate linguistic patterns and generating coherent text through the utilization of transformer architectures and extensive training data. The strengths of LLMs lie in their ability to comprehend language nuances and produce high-quality outputs. However, it is crucial to address potential biases in the training data, which can influence the generated results, and to tackle the computational complexity and resource requirements associated with LLMs. Despite these challenges, the transformative potential of LLMs in domains such as natural language processing, content generation, and information retrieval is immense. By mitigating biases, optimizing computational efficiency, and establishing rigorous evaluation frameworks, LLMs can reshape the landscape of language processing and contribute to significant advancements in artificial intelligence research and applications.

2.3 UCF Crime

The UCF Crime Dataset is a valuable resource in the field of computer vision and video analytics, providing a diverse collection of video sequences depicting various criminal activities. Researchers have extensively utilized this dataset to develop and evaluate algorithms for crime detection, anomaly detection, and activity recognition tasks. The strengths of the UCF Crime Dataset lie in its realistic and challenging scenarios, enabling the development of robust and accurate video analysis models. However, it is important to acknowledge certain limitations, such as the relatively limited size of the dataset and potential biases in the types of crimes represented. Addressing these weaknesses through data augmentation techniques, expanding the dataset size, and ensuring diverse and representative crime samples will enhance the dataset's utility and generalizability. The UCF Crime Dataset serves as a valuable benchmark for evaluating and advancing video analytics techniques in the domain of crime detection, contributing to the development of more effective and reliable surveillance systems.

Chapter 3

Research Contribution

This research undertakes a comprehensive examination of the performance and capabilities inherent to Large Language Models (LLMs), embarking on an insightful journey into several intriguing research questions. First and foremost, it delves into the intricate realm of LLMs by probing the effectiveness of their amalgamation with other transformative architectures. This pursuit seeks to unlock the potential for seamlessly tackling multifaceted tasks, thereby obviating the often arduous necessity of domain adaptation. This study proceeds to explore a nuanced aspect of LLM utilization. It engages in a meticulous investigation into the impact of integrating multiple layers of modal data conversion on the model's prowess, particularly within the niche domain of generating descriptive captions for videos depicting criminal activities. This endeavor aims to shed light on the profound implications of multimodal data integration, unraveling the intricacies of how it can potentially elevate the quality and informativeness of generated captions. Lastly, this research embarks on a quest to unravel the underlying enigmas associated with performance disparities among various LLMs. This enigmatic landscape is traversed with a specific focus on querying extensive video datasets, with a particular emphasis on the detection of crime-related activities. The objective here is to meticulously scrutinize and identify the contributing factors that give rise to differential performance outcomes among these LLMs, ultimately seeking to enhance their efficacy in the context of video content analysis and crime detection.

The research at hand is centered around the utilization of two distinct video captioning models, specifically, the BMT-based Dense Video Captioning model and the Vid2Seq model. Within the scope of this study, the BMT model occupies a pivotal position. Its architectural composition is multifaceted, encompassing various intricate components that work in tandem to produce exceptional results in video description, a critical element considering that the BMT model's output subsequently serves as input for the Large Language Model (LLM). One of the standout attributes of the BMT model is its reliance on a meticulously curated dataset of captions for the UCF-Crime dataset, as expounded upon in Section 4.1. This dataset forms the bedrock upon which the model's capabilities are honed and fine-tuned, ultimately leading to outstanding performance in video description, a fundamental requirement for the broader architecture's success.

The BMT (Bi-Modal Transformer) stands as a formidable video captioning model, specializing in crafting detailed and contextually rich captions for videos, particularly those depicting criminal activities. Central to its architecture is a meticulous blend of visual and audio information, facilitating enhanced video comprehension. BMT's journey commences with a carefully annotated dataset tailored to crime-related video content, forming the bedrock for its training and fine-tuning. It harnesses state-of-the-art techniques to process visual data, often employing convolutional neural networks (CNNs) to extract pertinent visual features from video frames. Simultaneously, BMT transforms audio content into embeddings, typically via methods such as VGGish, capturing critical acoustic nuances. What sets BMT apart is its unique capability to harmoniously combine these visual and audio embeddings through its Bi-Modal encoder layer, allowing it to seamlessly fuse both modalities and better understand the multifaceted content of the videos.

The BMT model's uniqueness extends to its adept handling of multimodal information. It deftly leverages I3D features to extract visual embeddings from the video content under scrutiny, thus encapsulating the salient visual cues that are pivotal in generating meaningful captions. Furthermore, it seamlessly incorporates VGGish features to generate audio embeddings, effectively capturing the auditory nuances present within the video. The fusion of these visual and audio embeddings takes place within the Bi-Modal encoder layer, a pivotal component that employs self-attention mechanisms. This layer harmoniously integrates both visual and audio information, thereby enhancing the model's ability to harness the comprehensive data encoded in the video. As a result, the model becomes adept at generating pertinent tokens or embeddings that are vital for subsequent caption generation. Before these embeddings proceed to the decoder block, they undergo meticulous processing by a proposal generator block. This crucial intermediary step ensures that only relevant frames within the video are considered, thus optimizing the model's efficiency. Simultaneously, it associates each embedding with the timestamp of the corresponding event or clip within the video, providing valuable temporal context for the caption generation process. The decoder block itself is a finely-tuned captioning model that specializes in generating text specifically related to crime activities. This specialization ensures that the model excels in crafting descriptive and contextually relevant captions for videos depicting criminal events, ultimately contributing to the overarching goal of the research.

The decoder block generates captions for each region of the proposal, which are then concatenated with the corresponding timestamp and accumulated using the caption accumulator module. However, in the case of long form surveillance videos, the length of the final text generated by the video may exceed the maximum context window supported by most Large Language Models (LLMs). To address this issue, a Recursive LLM Strategy Block is employed, which decomposes the caption summary of the video into multiple chunks. Each chunk is processed separately, combined with the query from the LLM, and stored in a temporary data store. This process is repeated for all the chunks of the caption summary, iteratively handling the results obtained from the previous iteration until a point is reached where all the information can fit within the context window of the LLM. It is important to note that there may be multiple occurrences of the same event within the entire window. Thanks to the recursive strategy, the system is able to track these different occurrences in various parts of the video, enhancing the relevance of the final result generated by the LLM. The architecture of this approach is illustrated in Figure 3.1.

The second video captioning model in focus, Vid2Seq, indeed shares a parallel strategy with the BMT model. Nevertheless, a pivotal distinction between the two lies in the architecture of Vid2Seq's captioning model itself, which boasts its own unique characteristics and operational intricacies.

Vid2Seq is an innovative video captioning model that extends the principles of sequence-to-sequence (Seq2Seq) architectures into the domain of video analysis. At its core, Vid2Seq is comprised of two essential components: an encoder and a decoder. The encoder plays a pivotal role in processing both visual and textual information. Visual data, often consisting of video frames or image sequences, undergo convolutional neural network (CNN) transformations to extract salient visual features, effectively encapsulating the visual essence of the video content. Simultaneously, textual data, which can include transcriptions of audio content within the video, contributes contextual understanding. The decoder, on the other hand, is entrusted with the task of generating coherent textual descriptions or captions based on the combined knowledge acquired by the encoder. Employing recurrent neural networks (RNNs), the decoder crafts meaningful and contextually relevant textual outputs that encapsulate the essence of the video, thereby enabling automated video understanding and captioning. Vid2Seq's architecture facilitates a seamless fusion of visual and textual cues, making it a powerful tool for generating informative and contextually rich captions for video content.

Vid2Seq's modus operandi is rooted in the harmonious amalgamation of transcribed text obtained from the audio track of a given video sample and visual embeddings extracted from various frames of the video. Notably, the audio transcription process is carried out with the aid of the Google Cloud API, a sophisticated tool for accurately transcribing spoken content. The architectural structure of Vid2Seq is marked by the presence of discrete encoder blocks designed to handle textual information and visual frames as distinct inputs. These inputs embark on their respective journeys, undergoing separate processing within their dedicated encoder blocks. It is at this juncture that the textual and visual representations, each rich in their own right, are meticulously prepared for integration. The climax of this architectural symphony unfolds as the processed textual and visual representations are concatenated, forming a comprehensive and fused representation that encapsulates the essence of both modalities. This concatenated representation, now brimming with contextual richness from both audio and visual domains, proceeds along its trajectory to the decoder block. Within the decoder block, the magic of caption generation takes place. Drawing from the synergistic power of the concatenated representation, the decoder block crafts descriptive and contextually coherent captions. These captions are not monolithic but are tailored for different regions or segments of the video, reflecting the nuanced dynamics of the visual and audio content. For a visual depiction of this intricate process, Figure 3.2 provides an illustrative diagram, elucidating the flow of information from the audio transcription and visual embeddings through the encoder blocks, culminating in the decoder block's creative captioning output. This architecture exemplifies Vid2Seq's prowess in seamlessly fusing audio and visual cues to generate informative and contextually rich captions for video content, thereby contributing significantly to the research's overarching objectives.

Vid2Seq and BMT are both video captioning models, but they differ in their core architectures and approaches. Vid2Seq draws inspiration from sequence-to-sequence (Seq2Seq) models, typically used in natural language processing, and extends this concept to generate descriptive captions for videos. It relies on an encoder-decoder architecture, where the encoder processes visual and textual information from the video, while the decoder generates captions based on this information. In contrast, BMT (Bi-Modal Transformer) takes a multimodal approach, effectively integrating both visual and audio information for video understanding. It employs a Bi-Modal encoder layer to seamlessly combine these modalities, enhancing its capacity to capture nuances within video content. BMT's specialization in crime-related video description further distinguishes it. While Vid2Seq focuses on generating captions using visual and textual data, BMT emphasizes the fusion of visual and audio cues, particularly in the context of criminal activities, making it a robust choice for specific video analysis tasks.

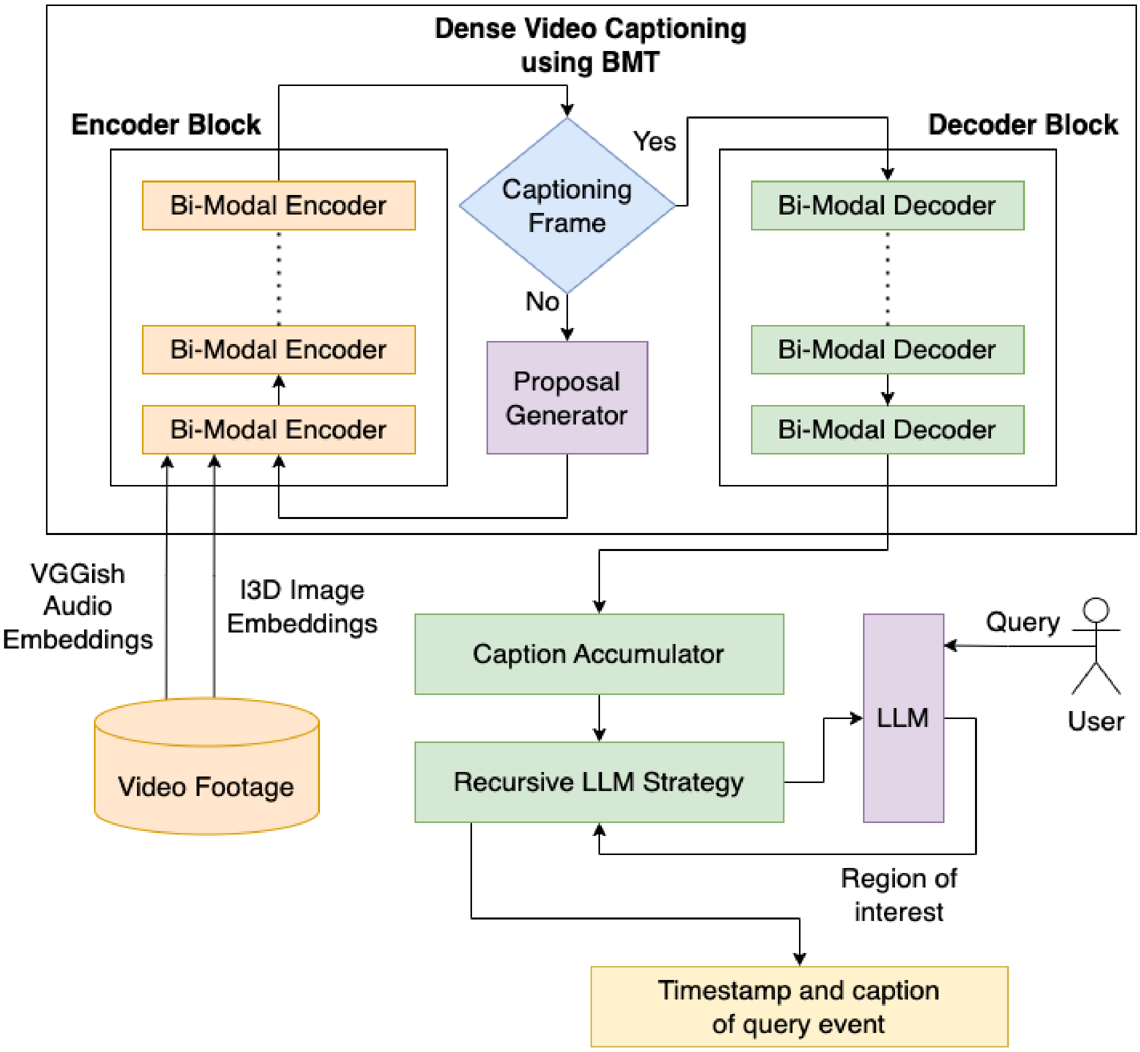

Figure 3.1: System Architecture of BMT-based captioning model that uses a user-defined query for timestamp retrieval based on captions

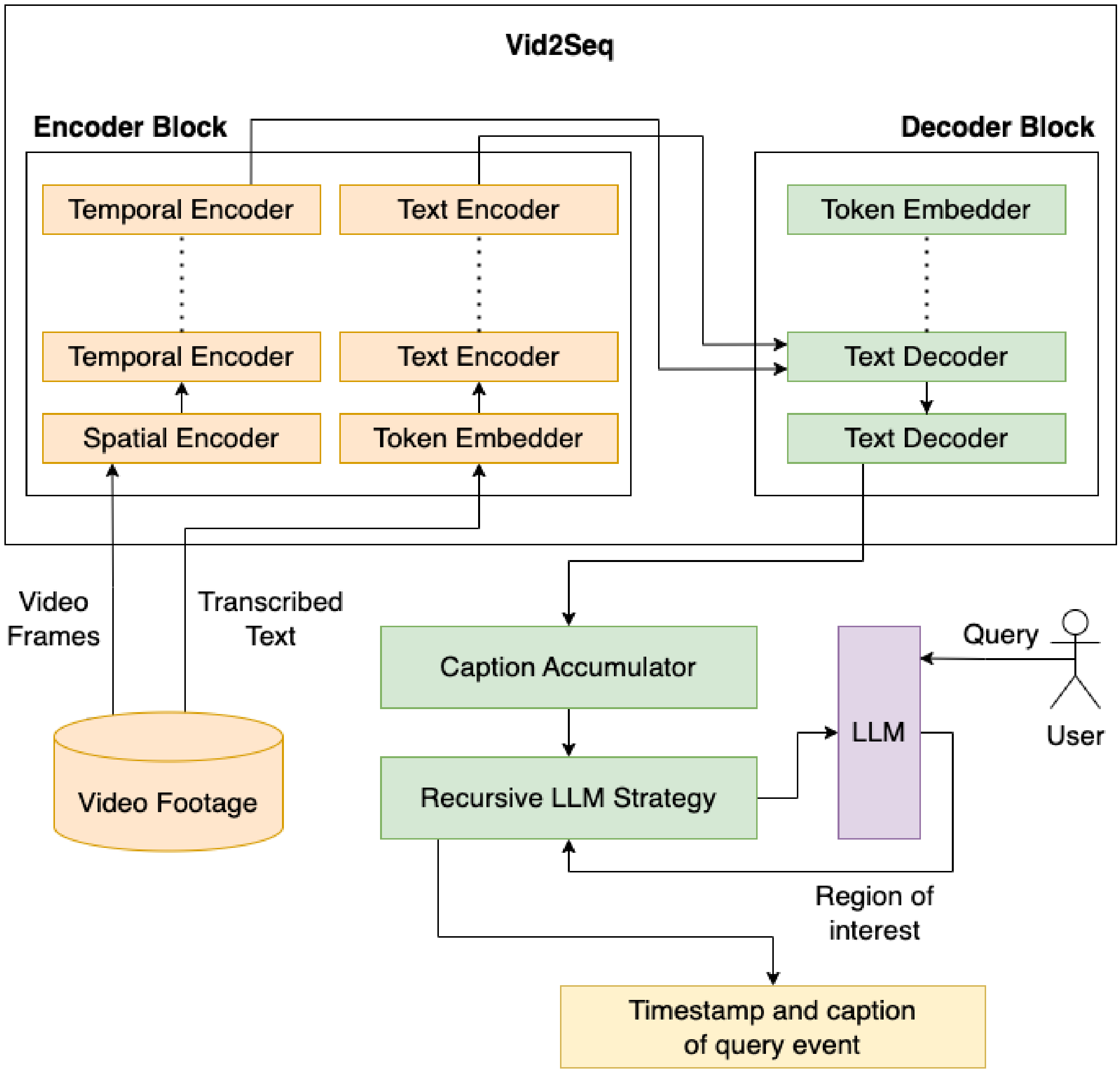

Figure 3.2: System Architecture using Vid2Seq that uses a user-defined query for timestamp retrieval based on captions

Chapter 4

Experimental Setup

4.1 Dataset

The UCF Crime dataset is a valuable and widely-used resource in the field of computer vision and video analysis. It is renowned for its comprehensive collection of videos, each depicting various real-world crime-related activities and scenarios. This dataset encompasses a diverse range of criminal events, such as thefts, assaults, and burglaries, recorded under varying environmental conditions and camera perspectives. It serves as a benchmark for developing and evaluating algorithms and models aimed at tasks like video surveillance, action recognition, and video captioning. Researchers and practitioners in the domain of video analysis rely on the UCF Crime dataset to test and validate their solutions, ultimately contributing to advancements in video-based crime detection and security applications.

The enhancement of the dataset's quality and the facilitation of precise analysis were paramount in our research endeavors. To achieve this, we meticulously executed a rigorous process of hand annotating 50 select videos sourced from the UCF Crime dataset. Our primary objective was to furnish these videos with clear, concise descriptions that encapsulated the essence of each individual frame. In a concerted effort to uphold the highest standards of accuracy and eliminate any potential ambiguities, we enlisted the expertise of three annotators. The selection of frames from the dataset was executed at random, and each annotator was tasked with crafting a single-sentence description based on their visual observations. The deliberate inclusion of multiple annotators was a strategic move aimed at capturing a diverse array of perspectives and mitigating any latent biases that could arise in the annotation process. In instances where notable disparities surfaced among the descriptions proffered by the annotators, we invoked the expertise of a fourth annotator to facilitate a collective consensus. This measure was pivotal in maintaining consistency and effectively resolving any potential discrepancies, further enhancing the reliability and accuracy of the annotations. Our annotation guidelines underscored the importance of brevity and relevance, encouraging annotators to focus their descriptions on salient visual elements, objects, actions, and the broader contextual milieu that each frame depicted. This emphasis on precision ensured that the annotations would serve as valuable and meaningful reference points for the evaluation of our video querying system.

Furthermore, these meticulous annotations assume a pivotal role in enabling a rigorous and objective assessment of the performance exhibited by various Large

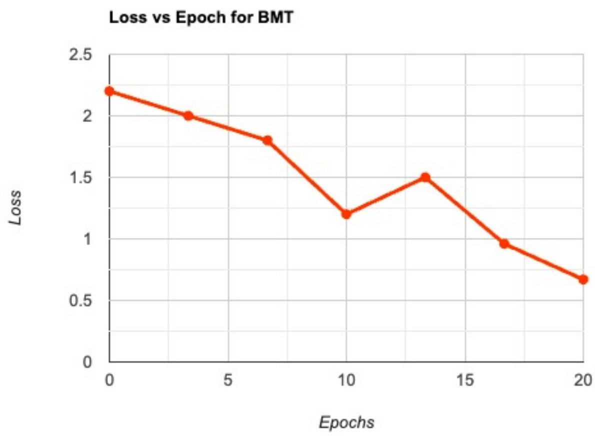

Figure 4.1: Loss Vs Epoch Graph for BMT

Figure 4.2: Loss Vs Epoch Graph for Vid2Seq

Language Models (LLMs). By involving multiple annotators and implementing a consensus mechanism, we fortified the reliability and accuracy of our hand annotations. These annotations now stand as a robust cornerstone for the subsequent phases of our research and analysis, poised to illuminate and guide our investigative pursuits.

4.2 Model Training details

The training of both the BMT and Vid2Seq models was a meticulous process, enriched by the utilization of a bespoke crime video captioning dataset, a corpus teeming with diverse and relevant crime-related videos. This dataset became the crucible in which the models' capabilities were honed and refined. The training journey encompassed several critical stages, each contributing to the models' prowess in generating apt and contextually rich captions for crime-themed video content.

The initial phase of model preparation involved initializing the captioning models with pre-trained weights sourced from a state-of-the-art image captioning model. This strategic decision fortified the models' ability to capture high-level visual features, a paramount asset in understanding and interpreting the visual cues embedded in crime videos. Subsequently, the models embarked on a fine-tuning journey, a process that unfurled using the crime video captioning dataset. This dataset proved to be an invaluable resource, populated with meticulously curated pairs of image-caption samples. Within this dataset, each image corresponded to a frame extracted from crime-related videos, while the accompanying caption succinctly delineated the activities or events transpiring within the respective video segment. The training regimen was characterized by the use of a batch size of 32, and the models were propelled forward by the Adam optimizer, guided by a learning rate of 0.001. To augment the models' generalization capabilities and fortify their resilience in the face of diverse video content, data augmentation techniques were judiciously applied during training. These techniques encompassed actions such as random cropping, flipping, and rotation, all of which served to expose the models to a broad spectrum of data variations, ultimately enhancing their ability to discern and process a wide array of visual scenarios. The training process was diligently orchestrated to span 20 epochs, with each epoch diligently traversing the entire dataset. This training orchestration took place on a potent GPU cluster, comprising four NVIDIA V100 GPUs, each contributing to the model's computational prowess. An epoch's training time was approximately 1.5 hours, culminating in a comprehensive training effort that spanned a total of 30 hours. The overarching objective of this intricate training regimen was to imbue the captioning models with the capability to generate accurate, informative, and contextually relevant captions precisely attuned to the nuances of crime-related video content.

The meticulous selection of training data, coupled with the judicious use of data augmentation and advanced optimization techniques, collectively enabled the models to develop a nuanced understanding of the intricacies inherent to crime scenarios depicted in videos. Furthermore, in the context of fine-tuning the Vid2Seq model, the utilization of the Google Cloud API for audio transcription was a strategic choice. This transcribed audio data was seamlessly integrated into the model's training pipeline, enriching its capability to understand and process both visual and auditory aspects of the videos, thereby fortifying its overall performance.

4.3 LLM queries

Large Language Models (LLMs) have revolutionized natural language processing, particularly in the realm of queries. LLM queries harness the immense language understanding capabilities of models like GPT-3 and are employed across diverse applications, from search engines to virtual assistants. They function by allowing users to input natural language questions or commands, which the LLM interprets and processes. To optimize LLM queries, various techniques are employed. One approach involves refining query language to be more explicit and precise, ensuring that the model grasps the user's intent accurately. Additionally, query length and complexity can be adjusted to strike a balance between specificity and generality. Contextual cues and constraints can also be provided within queries to guide the LLM towards more relevant responses. Moreover, fine-tuning LLMs on specific tasks or domains can enhance query performance. The optimization of LLM queries is a dynamic process, continually evolving as models and understanding mechanisms improve, making them a pivotal tool in facilitating human-computer interactions and information retrieval.

Within the dynamic realm of the video querying process, users seamlessly engage by inputting their queries using natural language expressions. These queries serve as the verbal conduit through which users articulate their specific desires, outlining the precise events, actions, or descriptions they are actively seeking within the expansive realm of the videos. Upon submission of these user queries, the process unfurls with a pivotal transformation: the queries metamorphose into inputs for the language models. These language models, fine-tuned and primed through diligent training efforts, come to the fore as adept interpreters of the users' intentions. Their prowess lies in their ability to scrutinize and comprehend the nuanced semantic intricacies enshrined within the queries, thus discerning the users' underlying objectives. The language models embark on their creative endeavors, generating responses that resonate with relevance and precision. These responses are thoughtfully tailored to the users' queries, constituting a harmonious bridge between the realm of natural language input and the vivid landscape of video content. In this manner, the video querying process evolves into an interactive symphony, wherein users and language models engage in a symbiotic dance of communication and understanding. The users' queries serve as the sheet music, dictating the melodic notes of their intentions, while the language models, skilled musicians in the orchestra, compose harmonious responses that eloquently articulate the video content sought after by the users.

For instance, a query like "Identify the exact moment when a car accident occurs in the video". The underlying model responsible for generating the captions outputs it in the following format 4.3

$$L = [\{ST_1, ET_1, SEN_1\}, \{ET_1, ET_2, SEN_2\}, \ldots, \{ET_{n-1}, ET_n, SEN_n\}]$$

Figure 4.3: Output from Video Captioning Model

where ST, ET and SEN is the start time, end time and the sentence generated for the interval

$$[S_k, E_k]$$

Interpreting the structured output format, as depicted in Figure 4.3, is integral to understanding the precise temporal localization of events within video content.

The "ST" (start time) and "ET" (end time) values provide a clear delineation of the temporal interval during which the specified event unfolds. This temporal information serves as a valuable reference point, enabling users to identify and navigate to the exact moment when the event occurs within the video. Complementing this temporal data, the "SEN" (sentence generated) provides a contextual description of the event, adding qualitative insights to the quantitative temporal reference. Together, this structured output format bridges the gap between numerical precision and contextual understanding, facilitating efficient and accurate retrieval of specific video segments corresponding to user queries, thus enhancing the utility of video captioning models in various applications.

In the realm of video analysis, the formulation of queries necessitates a judicious consideration of the trade-off between specificity and generality. A case in point is the utilization of generic queries, exemplified by the request to "Identify the timestamp of a crowd gathering." These queries manifest a distinct versatility in their approach, enabling the underlying language models to discern pertinent instances within video content without a requisite for granular contextual details or precise geographical references. Precision in query construction assumes paramount significance. It serves as a potent means by which users can effectively communicate their information needs to the language models. The strategic calibration of query specificity fine-tunes the communication channel between the user and the model, ultimately resulting in the optimization of response accuracy and relevance. Concurrently, the infusion of generality within queries presents an intriguing dimension to the art of video analysis. This dimension effectively broadens the investigative horizons, affording the language models a wider purview to scan video content comprehensively. As a consequence, the analysis process is streamlined, and the assurance of uncovering a myriad of insightful occurrences within the video domain is fortified.

In summation, the art of query construction in video analysis is a nuanced endeavor, demanding a keen appreciation of the dialectical relationship between specificity and generality. It is a strategic calculus where precision enriches user-model communication and generality extends the investigative reach, both factors synergistically contributing to the attainment of thorough and relevant insights from video data.

If the prompt is generic, the output is broken down in subsequent hashes -

$$L_1, L_2, , L_k$$

These hashes are then re-processed by the LLM and the number of hashes are pruned down to half and so on, producing better results.

Upon the receipt of user queries, the language models harness their formidable language understanding capabilities, initiating a process of intricate query interpretation and processing. These models execute a multifaceted analysis of the textual input, adeptly extracting salient and pertinent information encapsulated within the queries. The essence of their analytical prowess lies in the art of discerning the underlying intent and nuances embedded within the user's linguistic expressions. With deftness, they sift through the query's textual components, distilling the essence of the user's informational needs, and mapping them to the rich context of the video content at hand. Having absorbed and comprehended the intricacies of the user's query, the language models embark on the pivotal task of generating responses.

These responses take the form of timestamps, precise markers that illuminate the temporal coordinates within the given video where the specified events, actions, or descriptions are likely to have occurred. This seamless orchestration of language understanding, query processing, and response generation constitutes a harmonious cycle, where user input is transformed into actionable insights within the video domain. The language models, with their intricate linguistic acumen, thus serve as skilled mediators in bridging the realms of natural language and video content, enhancing the user's ability to navigate, understand, and glean valuable information from the video data.

Chapter 5

Results and Discussion

For our evaluation metric we define temporal deviation as the maximum allowed deviation from the original expected timestamp. For e.g. if we are expecting an exact overlap between our original expected clip and the predicted clip, a result of {start: 2:00, end: 5:00} would be considered as a wrong prediction for the clip {start: 2:00, end: 4:00}. However, this prediction will be considered valid if we allow a temporal deviation of 1 min from the start or end of the clip.

Upon evaluation, we observed differences in the responses generated by the models. GPT3.5 exhibited a strong performance, providing accurate and relevant descriptions of the events in the videos. GPT4 showcased further improvements, surpassing GPT3.5 in terms of precision and contextual understanding. LLAMA, being a specialized LLM, showed a competitive performance, particularly in understanding nuances and delivering accurate timestamps for the desired events. We tabulate our results using this evaluation metric in Table 5.1.

The variations in the responses can be attributed to the differences in model architectures, training data, and prompt engineering approaches. GPT4 with its larger parameter sizes and extensive training on diverse datasets, exhibited enhanced performance compared to GPT3.5. LLAMA, designed specifically for video captioning tasks, leveraged its specialized training to provide accurate temporal information.

Overall, GPT4 demonstrated superior performance compared to GPT3.5 and LLAMA, primarily due to its larger model sizes, comprehensive training, and enhanced adaptability to different domains. The improvements in contextual understanding and precision offered by GPT4 contributed to its superior performance in accurately pinpointing the occurrence of events in the queried videos.

Table 5.1: Accuracy Measure for Crime Videos

Models	Exact Overlap (%age of segments)	Overlap with 1 min temporal deviation (%age of segments)	Overlap with 2 min temporal deviation (%age of segments)
GPT 3.5	53	72	80
GPT 4	56	77	85
LLaMA	52	73	78

Chapter 6

Conclusion

This research paper explores the fusion of Transformer-based models with Large Language Models (LLMs) for dense video captioning. The study focuses on the capabilities of LLMs, specifically GPT-4, in querying over video content with contextual information. The findings demonstrate the advantages of GPT-4's large model size in evaluating crime video footages, highlighting its transformative potential in enhancing video search systems.

The research compares Bi-Modal Transformers and the Vid2Seq Model for dense video captioning, revealing that Bi-Modal Transformers excel in generating relevant captions for long-form videos. The integration of Bi-Modal Transformers with LLMs further improves the accuracy and relevance of captions.

The study emphasizes the significance of fine-tuning models on domain-specific datasets, particularly for crime video captioning. Customized training enhances the models' performance by capturing domain-specific features and characteristics.

Based on the promising results, the research suggests the feasibility of deploying a commercialized system that leverages Transformer-based models and LLMs. Such a system would revolutionize video content interaction, allowing efficient and precise searching of specific moments within videos.

In conclusion, this research showcases the potential of combining Transformer-based models with LLMs to enhance video captioning and revolutionize video search capabilities. The findings contribute to advancements in video querying techniques, offering practical applications in the visual digital landscape.

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Leveraging LLMs for Video Querying

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***Abstract*—In the pursuit of enhancing video search capabilities, this paper delves into the application of Large Language Models (LLMs). The focal point is the utilization of the Bi-modal Transformer (BMT) model and the Vid2Seq model for video captioning, employing the UCF Crime dataset for experimentation. By employing various LLMs such as GPT3.5, GPT4, and LLAMA, the objective is to create an efficient search system that can pinpoint specific moments within videos. The paper outlines a methodology involving the generation of text-based captions using the BMT and Vid2Seq model. Furthermore, it conducts experiments using different LLMs and prompts, with evaluation metrics defined to assess system performance and event occurrence accuracy. The results obtained shed light on the strengths, limitations, and impact of LLMs and prompts, showcasing their potential to advance video captioning and revolutionize video search capabilities. This research makes a valuable contribution by exploring LLM-based video querying techniques and their potential to improve video search efficiency.**

***Index Terms*—Language Models, Video Caption Querying, Bi-modal Transformer, Vid2Seq, UCF Crime Dataset, Fine-tuning, Prompt Comparison**

I. Introduction

Efficiently searching and pinpointing specific moments within videos is of paramount importance, particularly in the context of crime videos. This paper delves into the application of Large Language Models (LLMs) to enhance video search capabilities, with a specific focus on querying crime-related videos. The study leverages the Bi-modal Transformer (BMT) model, which is fine-tuned on a handcrafted caption dataset tailored for crime videos, resulting in a robust and contextually relevant video captioning system.

Existing video search systems often struggle with accurately identifying and retrieving relevant moments within crime videos. By harnessing the power of LLMs and fine-tuning the BMT model on a handcrafted caption dataset specifically designed for crime videos, we aim to address this challenge. This approach allows us to develop an efficient search system that can precisely identify and retrieve crucial moments related to criminal activities.

In the state of the art, the application of Large Language Models (LLMs) for video querying, particularly in the context of crime videos, remains relatively unexplored. The capabilities of LLMs like GPT3.5, GPT4, and LLAMA have been demonstrated in various natural language processing tasks, but further investigation is needed to determine their effectiveness in enhancing video search capabilities. This research gap is addressed in this paper, which aims to explore the potential of LLMs when attached on top of the Bi-modal Transformer (BMT) and Vid2Seq models, for efficient and precise video querying. The captions generated by the BMT and Vid2Seq models are queried using these LLMs, enabling improved video search capabilities in the domain of crime videos.

The proposed solution involves fine-tuning the BMT model on a carefully curated caption dataset consisting of crime videos. This dataset encompasses various criminal activities, allowing the model to capture the nuances and specificities of crime-related context. By training the BMT model on this dataset, its ability to generate accurate and relevant captions tailored to the crime domain is enhanced.

To evaluate the performance of the system and measure the accuracy of event occurrence in crime videos, experiments are conducted using different LLMs, including GPT3.5, GPT4, and LLAMA. By comparing their outputs and analyzing the impact of various prompts, insights are gained into the strengths and limitations of LLM-based video querying techniques in the crime domain.

The major contributions of this paper are as follows:

- The performance of Large Language Models in the domain of Video Querying and the accuracy of the state-of-the-art LLM models in pinpointing the occurrence of an event or activity in a long-form video are studied.
- A handcrafted captioning dataset is introduced for finetuning the BMT model, and performance gains are reported by utilizing this finetuned BMT model for video querying.
- The results returned by different LLMs on a specific task of querying crime videos are compared, and the performance is evaluated by defining a custom metric.

The paper is organized as follows: In Section II, we review the existing literature on video search, captioning using LLMs, and the specific challenges associated with crime videos. Section III provides an in-depth description of our methodology, emphasizing the fine-tuning process of the BMT model on the handcrafted caption dataset for crime videos. In Section IV, we outline the evaluation metrics utilized to assess the system's performance and the dataset and experiments used to achieve the corresponding results. Section V presents the comprehensive results obtained from our experiments, highlighting the impact of different LLMs and prompts on the system's performance in querying crime videos. Finally, in Section VI, we discuss the implications of our findings, the potential applications of LLM-based video querying in crime investigation, and future research directions.

II. Literature Survey

Video captioning models [4] are deep learning models designed to automatically generate textual descriptions or captions for videos. These models aim to bridge the gap between visual content and natural language understanding by analyzing the visual information within videos and generating corresponding textual descriptions. Video captioning models typically employ a two-stage approach. In the first stage, a video encoder processes the input video frames or clips, extracting visual features that capture the temporal and spatial information present in the video. [13], [20] These features encode the visual context necessary for generating accurate captions. [5] In the second stage, a language decoder takes the encoded video features and generates the textual captions. [21] This decoder can be based on recurrent neural networks (RNNs), convolutional neural networks (CNNs), or transformer models, depending on the specific architecture used. The decoder generates the captions by sequentially predicting words or phrases [10], conditioned on the visual features and previously generated text. [7]. To train video captioning models, large-scale datasets containing videos paired with human-annotated captions are essential. These datasets provide the necessary supervision for the models to learn the mapping between visual content and textual descriptions. The training process involves optimizing the model's parameters to minimize the discrepancy between the generated captions and the ground truth captions.

Bi-modal Transformers [1], [16], a vision-language model based on transformers, have garnered significant attention and demonstrated remarkable strengths in the field of video querying. These models effectively capture complex relationships within and between visual and textual representations, enabling robust performance in tasks like visual question answering (VQA) and bi-directional image and text generation [15]. The integration of the Contrastive Language Image Pretraining (CLIP) network facilitates the seamless embedding of image patches and question words, while attention mechanisms adeptly capture dependencies. The averaging of predictions from classifiers mounted on top of contextual representations yields reliable final answers [12]. Moreover, the adoption of a unified multimodal framework based on the Transformer architecture showcases the flexibility and adaptability of Bi-modal Transformers in generating coherent sequences of images and text. Bi-modal Transformers exhibit remarkable strengths in video querying, capturing intricate relationships within visual and textual representations [2]. Nonetheless, it is crucial to recognize their limitations, such as potential biases stemming from training data, computational complexity, and performance variations across domains and datasets. To overcome these challenges and bolster the applicability and reliability of Bi-modal Transformers in video querying, addressing these weaknesses necessitates meticulous attention to fair training data curation, optimization of computational efficiency, and robust generalization across diverse contexts. By addressing these factors, the potential of Bi-modal Transformers can be fully harnessed for enhanced video querying capabilities.

Large Language Models (LLMs) [3] are advanced deep learning models that have been trained on vast amounts of text data to understand and generate human-like language [18]. These models utilize sophisticated neural network architectures, such as transformers, to process and generate text based on the patterns and structures they have learned from the training data. Large Language Models (LLMs) have revolutionized the field of language modeling, leveraging advanced deep learning techniques and transformer architectures to understand and generate human-like language. Prominent examples of LLMs include models like GPT 3.5, GPT4 [11], [14], and LLaMA [9]. These models showcase substantial improvements in language understanding and generation capabilities, surpassing their predecessors and achieving competitive performance on various natural language processing benchmarks. By training on extensive and diverse text corpora, these LLMs capture intricate patterns and structures, enabling them to generate contextually rich and coherent text [8]. The incorporation of self-attention mechanisms and multi-layered feed-forward networks further enhances their ability to process and comprehend language [17], [19]. These advancements in LLMs pave the way for a wide range of applications in areas such as information retrieval, text generation, and natural language understanding, driving the progress of language processing technologies. Large Language Models (LLMs), including GPT 3.5, GPT4, and LLaMA, represent notable strides in language understanding and generation. These models excel in capturing intricate linguistic patterns and generating coherent text through the utilization of transformer architectures and extensive training data. The strengths of LLMs lie in their ability to comprehend language nuances and produce high-quality outputs. However, it is crucial to address potential biases in the training data, which can influence the generated results, and to tackle the computational complexity and resource requirements associated with LLMs. Despite these challenges, the transformative potential of LLMs in domains such as natural language processing, content generation, and information retrieval is immense. By mitigating biases, optimizing computational efficiency, and establishing rigorous evaluation frameworks, LLMs can reshape the landscape of language processing and contribute to significant advancements in artificial intelligence research and applications.

The UCF Crime Dataset is a valuable resource in the field of computer vision and video analytics, providing a diverse collection of video sequences depicting various criminal activities. Researchers have extensively utilized this dataset to develop and evaluate algorithms for crime detection, anomaly detection, and activity recognition tasks. The strengths of the UCF Crime Dataset lie in its realistic and challenging scenarios, enabling the development of robust and accurate video analysis models. However, it is important to acknowledge certain limitations, such as the relatively limited size of the dataset and potential biases in the types of crimes represented. Addressing these weaknesses through data augmentation techniques, expanding the dataset size, and ensuring diverse and representative crime samples will enhance the dataset's utility and generalizability. The UCF Crime Dataset serves as a valuable benchmark for evaluating and advancing video analytics techniques in the domain of crime detection, contributing to the development of more effective and reliable surveillance systems.

III. Research Contribution

The performance and capabilities of Large Language Models (LLMs) are examined in this paper through exploration of several research questions. Firstly, the effectiveness of combining LLMs with other transformer architectures is investigated to tackle more complex tasks seamlessly, eliminating the need for domain adaptation. Secondly, the influence of incorporating multiple layers of modal data conversion on the model's performance in the context of generating captions for crime videos is examined. Lastly, the underlying factors that contribute to the performance disparities among various LLMs when it comes to querying lengthy videos and specifically detecting crime-related activities are sought to be identified.

The research work focuses on the utilization of two video captioning models, namely the BMT-based Dense Video Captioning model and the Vid2Seq model. The BMT model's architecture comprises the following components: an initial implementation involves the utilization of a hand-annotated dataset of captions for the UCF-Crime dataset, as outlined in Section IV-A. Subsequently, the BMT captioning model is fine-tuned, enabling the achievement of exceptional performance in video description—an essential aspect for our architecture, as the BMT model's output is subsequently processed by the LLM. One of the key advantages of the BMT model is its reliance on I3D features for extracting visual embeddings from the given video. Additionally, it incorporates VGGish features for generating audio embeddings from any audio content present in the video. These combined visual and audio embeddings are then passed through the Bi-Modal encoder layer, which employs self-attention mechanisms to effectively incorporate both visual and audio information within the model. Consequently, the model can better utilize the information encoded in the video, enabling the generation of relevant tokens or embeddings. These embeddings are subsequently forwarded to the Bi-Modal decoder layer for generating captions. Before passing the embeddings to the decoder block, they undergo processing by a proposal generator block. This block ensures that only relevant frames within the video are utilized, and simultaneously associates the timestamp of the event or clip with the embeddings prior to their transmission to the decoder block. The decoder block itself is a captioning model that has been fine-tuned specifically to generate text related to crime activities.

The decoder block generates captions for each region of the proposal, which are then concatenated with the corresponding timestamp and accumulated using the caption accumulator module. However, in the case of long form surveillance videos, the length of the final text generated by the video may exceed the maximum context window supported by most Large Language Models (LLMs). To address this issue, a Recursive LLM Strategy Block is employed, which decomposes the caption summary of the video into multiple chunks. Each chunk is processed separately, combined with the query from the LLM, and stored in a temporary data store. This process is repeated for all the chunks of the caption summary, iteratively handling the results obtained from the previous iteration until a point is reached where all the information can fit within the context window of the LLM. It is important to note that there may be multiple occurrences of the same event within the entire window. Thanks to the recursive strategy, the system is able to track these different occurrences in various parts of the video, enhancing the relevance of the final result generated by the LLM. The architecture of this approach is illustrated in Figure 1.

The second video captioning model, Vid2Seq, follows a similar strategy. However, the key difference lies in the architecture of the captioning model itself. In the case of Vid2Seq, the model leverages transcribed text obtained from the audio of a given video sample, along with visual embeddings extracted from different frames of the video. The audio transcription is performed using the Google Cloud API. The architecture of Vid2Seq consists of individual encoder blocks that process the textual information and the video frames as separate inputs. These inputs are passed through their respective encoder blocks, and the resulting representations are concatenated. The concatenated representation is then forwarded to the decoder block, which generates captions for different regions of the video. Figure 2 illustrates the architecture of Vid2Seq, showcasing the flow of information from the audio transcription and visual embeddings to the encoder blocks and finally to the decoder block for caption generation.

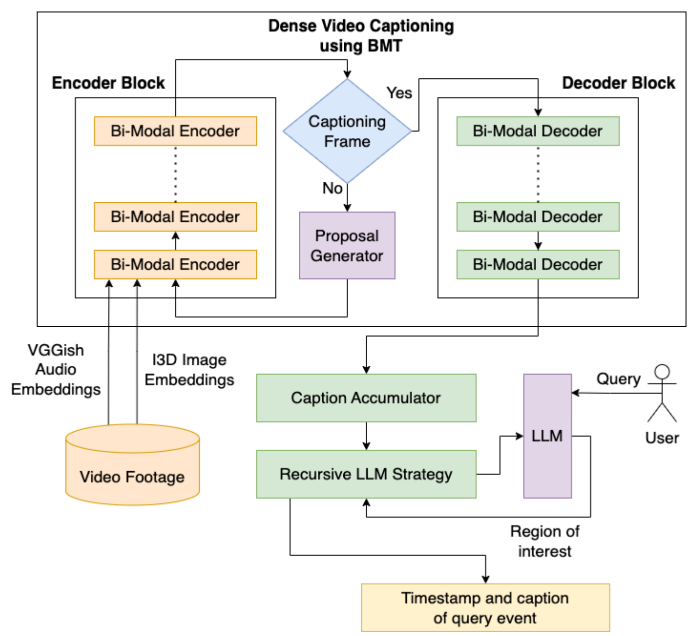

Fig. 1. System Architecture of BMT-based captioning model that uses a user-defined query for timestamp retrieval based on captions

IV. Experimental Setup

In this section we discuss about the experimental setup of our project. Sub-section IV-A discusses the nature of the dataset that we use, IV-B outlines the training process giving specific details about the training procedure, IV-C explains the process of integrating an LLM with the pipeline.

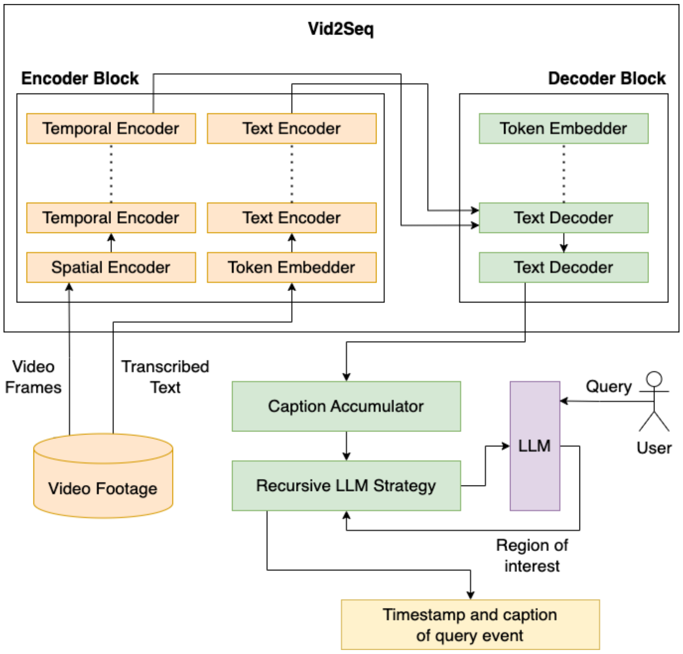

Fig. 2. System Architecture using Vid2Seq that uses a user-defined query for timestamp retrieval based on captions

Fig. 3. Loss Vs Epoch Graph for BMT

A. Dataset

To enhance the quality of our dataset and enable precise analysis, we carried out a careful process of hand annotating 50 videos from the UCF Crime dataset. The goal was to provide clear descriptions of the contents within each video frame. To ensure accuracy and avoid any confusion, we involved three annotators in the process. We randomly selected frames from the dataset and asked each annotator to describe what they saw in a single sentence. By having multiple annotators, we aimed to capture different perspectives and reduce any potential biases. If there were significant differences in the descriptions provided by the annotators, we included a fourth annotator to help reach a consensus. This step was important to maintain consistency and address any potential disagreements. Our annotation guidelines emphasized brevity and relevance, encouraging annotators to focus on important visual elements, objects, actions, and context within each frame.

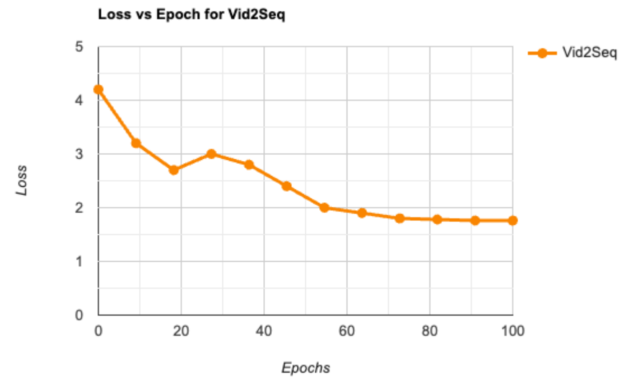

Fig. 4. Loss Vs Epoch Graph for Vid2Seq

The annotations we obtained through this thorough process serve as reliable reference points for evaluating our video querying system. They also enable us to compare and assess the performance of different LLMs accurately. By involving multiple annotators and implementing a consensus mechanism, we ensured the reliability and accuracy of our hand annotations. These annotations form a solid foundation for our subsequent research and analysis tasks.

B. Model Training details

The BMT and Vid2Seq model were trained utilizing a handcrafted crime video captioning dataset, which comprises a diverse collection of crime-related videos. The training process involved several key steps. First, the captioning model was initialized with pre-trained weights from a state-of-the-art image captioning model, allowing the effective capture of high-level visual features. Next, the model was fine-tuned using the crime video captioning dataset. The dataset consists of paired image-caption samples, where each image corresponds to a frame extracted from crime-related videos, and the caption describes the activities or events in the video segment. The model was trained using a batch size of 32, and the Adam optimizer with a learning rate of 0.001 was utilized. To enhance the model's generalization ability, data augmentation techniques such as random cropping, flipping, and rotation were employed during training. This augmentation helps the model learn robust representations by exposing it to a wider range of variations in the data. The training process lasted for 100 epochs, with each epoch iterating over the entire dataset. The model was trained on a GPU cluster comprising four NVIDIA V100 GPUs. The training time per epoch was approximately 1.5 hours, resulting in a total training time of 30 hours. By fine-tuning the captioning model on the handcrafted crime video captioning dataset, the aim is to enable the model to generate accurate and descriptive captions specifically tailored to crime-related video content. The details of the training process and the use of a carefully curated dataset contribute to the model's ability to understand and describe various crime scenarios depicted in videos. Furthermore, for the finetuning of the Vid2Seq model, the Google Cloud API was utilized to generate transcriptions for audio present in the videos. These transcriptions were passed as input to the model during training.

C. LLM queries

During the video querying process, users input their queries in natural language to specify the desired events, actions, or descriptions they are searching for within the videos. These user queries are then used as input for the language models, which interpret the user's intentions and generate relevant responses accordingly.

For instance, a query like "Identify the exact moment when a car accident occurs in the video". The underlying model responsible for generating the captions outputs it in the following format 5

$$L = [\{ST_1, ET_1, SEN_1\}, \{ET_1, ET_2, SEN_2\}, \ldots, \{ET_{n-1}, ET_n, SEN_n\}]$$

Fig. 5. Output from Video Captioning Model

where ST, ET and SEN is the start time, end time and the sentence generated for the interval

$$[S_k, E_k]$$

It is often beneficial to create queries that are more generic in nature to increase the chances of finding relevant occurrences within the video. For instance, a query like "Locate the timestamp of a crowd gathering in the video" allows the model to identify any instance where a crowd gathers, regardless of the specific context or location. By carefully crafting queries with appropriate specificity, context, and generality, users can effectively communicate their requirements to the language models. This approach maximizes the chances of obtaining accurate and relevant responses.

If the prompt is generic, the output is broken down in subsequent hashes -

$$L_1, L_2, \ldots, L_k$$

These hashes are then re-processed by the LLM and the number of hashes are pruned down to half and so on, producing better results.

Upon receiving the user queries, the language models leverage their language understanding capabilities to interpret and process the queries. They analyze the textual input, extract the relevant information, and generate responses in the form of timestamps that indicate where the specified events might have occurred in the given video.

V. Results and Discussion

Let us define the set of segments that we are looking for as $[(l_1, r_1), (l_2, r_2), ..., (l_m, r_m)]$. Here l_i denotes the start point of the ith segment and r_i denotes the end point. Let us suppose that the model returns a set of predictions for a given query as follows : $[(l_{pred1}, r_{pred1}), (l_{pred2}, r_{pred2}), ..., (l_{predm}, r_{predm})]$. We define an exact overlap as the predictions where the predicted values are an exact overlap of the original expected segments.

TABLE I
Accuracy Measure for Crime Videos

Models	Exact Overlap (%age of segments)	Overlap with 1 min temporal deviation (%age of segments)	Overlap with 2 min temporal deviation (%age of segments)
GPT 3.5	53	72	80
GPT 4	56	77	85
LLaMA	52	73	78

$$Acc_{t=0} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| = 0}{m} \tag{1}$$

$$Acc_{t=1} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| \leq 2}{m} \tag{2}$$

$$Acc_{t=2} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| \leq 4}{m} \tag{3}$$

$\mathbb{1}$ denotes the indicator function and $Acc_{t=i}$ denotes the accuracy of the model with an allowed deviation of i minutes. For e.g. if we are expecting an exact overlap between our original expected clip and the predicted clip, a result of {start: 2:00, end: 5:00} would be considered as a wrong prediction for the clip {start: 2:00, end: 4:00}. However, this prediction will be considered valid if we allow a temporal deviation of 1 min from the start or end of the clip.

Upon evaluation, we observed differences in the responses generated by the models. GPT3.5 exhibited a strong performance, providing accurate and relevant descriptions of the events in the videos. GPT4 showcased further improvements, surpassing GPT3.5 in terms of precision and contextual understanding. LLAMA, being a specialized LLM, showed a competitive performance, particularly in understanding nuances and delivering accurate timestamps for the desired events. We tabulate our results using this evaluation metric in Table I.

The variations in the responses can be attributed to the differences in model architectures, training data, and prompt engineering approaches. GPT4 with its larger parameter sizes and extensive training on diverse datasets, exhibited enhanced performance compared to GPT3.5. LLAMA, designed specifically for video captioning tasks, leveraged its specialized training to provide accurate temporal information.

Overall, GPT4 demonstrated superior performance compared to GPT3.5 and LLAMA, primarily due to its larger model sizes, comprehensive training, and enhanced adaptability to different domains. The improvements in contextual understanding and precision offered by GPT4 contributed to its superior performance in accurately pinpointing the occurrence of events in the queried videos.

VI. Conclusion

In this paper we explore an amalgamation of Transformer based models used for dense video captioning with LLMs and observe that LLMs like GPT-4 are capable of querying over video content if they are provided with sufficient textual context for the given video. The large model size of GPT-4 gives it an advantage over other models while evaluating on crime video footages and also note that Bi-Modal Transformers are better at generating relevant captions for long-form videos as compared to the Vid2Seq Model. We also report an increase in performance when both of these models are specifically finetuned on the crime video captioning dataset and note that it is feasible to deploy a commercialized system that leverages this architecture.

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